EVALUATION

I Multiple Choice Questions

1. Suppose an alpha particle accelerated by a potential of V volt is allowed to collide with a nucleus of atomic number Z, then the distance of closest approach of alpha particle to the nucleus is

(a) $14.4\frac{Z}{V}$ Å (b) $14.4\frac{V}{Z}$ Å

(c) $1.44\frac{Z}{V}$ Å (d) $1.44\frac{V}{Z}$ Å

2. In a hydrogen atom, the electron revolving in the fourth orbit, has angular momentum equal to

(a) h (b) $\frac{h}{\pi}$

(c) $\frac{4h}{\pi}$ (d) $\frac{2h}{\pi}$

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3. Atomic number of H-like atom with ionization potential 122.4 V for $n = 1$ is

(a) 1 (b) 2 (c) 3 (d) 4

4. The ratio between the radius of first three orbits of hydrogen atom is

(a) 1:2:3 (b) 2:4:6

(c) 1:4:9 (d) 1:3:5

5. The charge of cathode rays particle is

(a) positive (b) negative

(c) neutral (d) not defined

6. In J.J. Thomson e/m experiment, electrons are accelerated through 2.6 kV enter the region of crossed electric field and magnetic field of strength 3.0×10^4 Vm^{-1} and 1.0×10^{-3} T, respectively, and pass through it and undeflected, then the specific charge is

(a) 1.6×10^{10} Ckg^{-1}

(b) 1.7×10^{11} Ckg^{-1}

(c) 1.5×10^{11} Ckg^{-1}

(d) 1.8×10^{11} Ckg^{-1}

7. The ratio of the wavelengths radiation emitted for the transition from $n = 2$ to $n = 1$ in Li^{++}, He^{+} and H is

(a) 1: 2: 3 (b) 1: 4: 9

(c) 3:2:1 (d) 4: 9: 36

8. The electric potential of an electron is given by $V = V_0 \ln\left(\frac{r}{r_0}\right)$, where r_0 is a constant. If Bohr atom model is valid, then variation of radius of nth orbit r_n with the principal quantum number n is

(a) $r_n \propto \frac{1}{n}$ (b) $r_n \propto n$

(c) $r_n \propto \frac{1}{n^2}$ (d) $r_n \propto n^2$

9. If the nuclear radius of ^{27}Al is 3.6 fermi, the approximate nuclear radius of ^{64}Cu in fermi is

(a) 2.4 (b) 1.2

(c) 4.8 (d)3.6

10. The nucleus is approximately spherical in shape. Then the surface area of nucleus having mass number A varies as

(a) $A^{2/3}$ (b) $A^{4/3}$

(c) $A^{1/3}$ (d) $A^{5/3}$

11. The mass of a $^{7}_{3}Li$ nucleus is 0.042 u less than the sum of the masses of all its nucleons. The average binding energy per nucleon of $^{7}_{3}Li$ nucleus is nearly

(a) 46 MeV (b) 5.6 MeV

(c) 3.9MeV (d)23 MeV

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